

Ch-3

Chemical kinetics

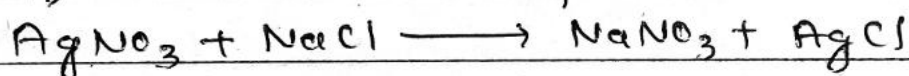
→ The branch of physical chemistry which deals with rate of chemical reaction is called chemical kinetics.

Types of reaction:-

i) Fast reaction

→ This type of reaction completes within fraction of time.

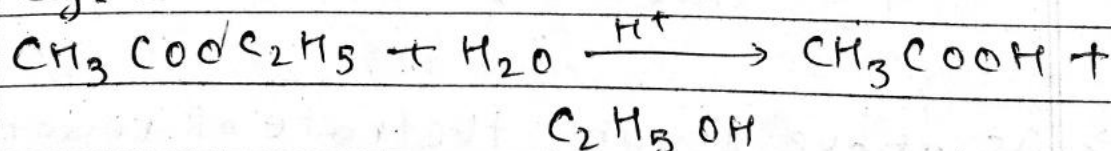
Eg. Ionic rxn in aqueous form.



ii) Moderate reaction

→ This type of reaction completes within measurable time (not very slow or fast)

Eg. Covalent bond rxns.



iii) Slow reaction

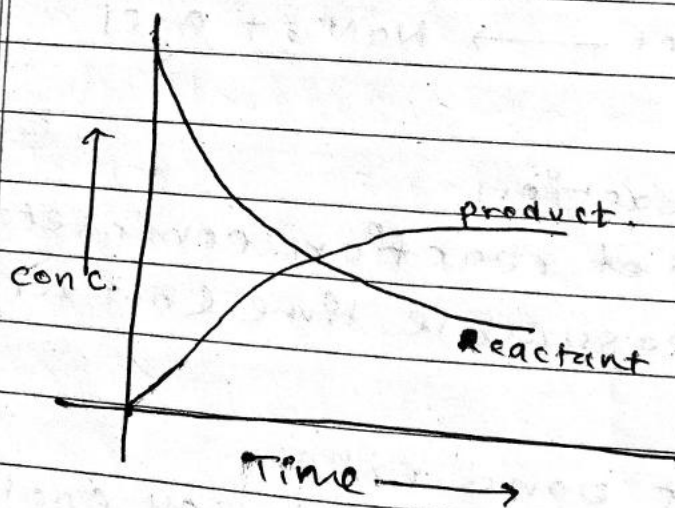
- This reaction completes in very long period of time and immeasurable.
- Eg. Rusting of iron.

### (\*) Rate of reaction.

- The rate of change of concentration of reactant or product in a reaction is called rate of reaction.

Rate of Reaction =  $\frac{\text{change in concentration of reactant or product}}{\text{time taken.}}$

It's unit is  $\text{mol Lit}^{-1} \text{time}^{-1}$ .



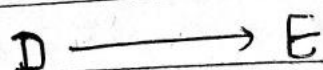
- As above graph, the rate of reaction is high at beginning and slowly decreases over time.
- So rate of reaction is not constant.

## Types of Rate of Reaction.

### i) Average rate of reaction

→ The rate of change in concentration of reactant or product over a certain period of time is called average rate of reaction.

Consider a reaction,



Let at time ' $t_1$ ', the concentration of D and E is  $[D_1]$  and  $[E_1]$  respectively.

at time ' $t_2$ ', the concentration of D and E is  $[D_2]$  and  $[E_2]$  respectively.

Now,

$$\text{average rate of rxn} = \frac{[D_2] - [D_1]}{t_2 - t_1} = \frac{[E_2] - [E_1]}{t_2 - t_1}$$

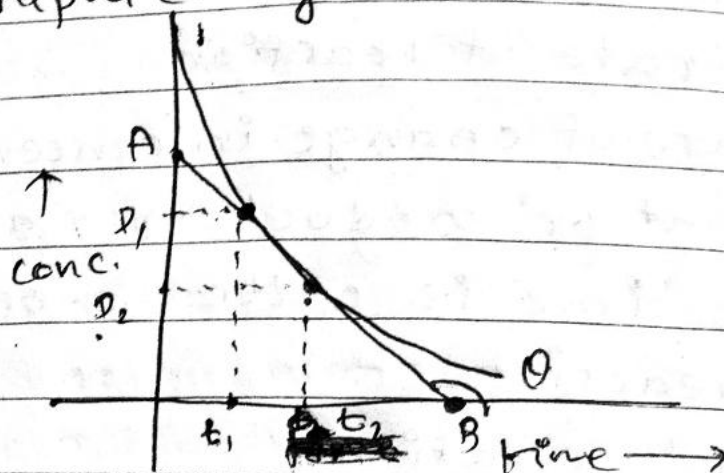
$$\text{suppose, } [D_2] - [D_1] = \Delta D$$

$$[E_2] - [E_1] = \Delta E$$

$$\therefore \text{average rate of rxn} = \frac{-\Delta D}{\Delta t} = \frac{\Delta E}{\Delta t}$$

In term of reactant, -ve sign is introduced because  $\Delta D$  is -ve ( $[D_2] < [D_1]$ ) and rate of reaction is always +ve.

Graphically.



Slope of line AB =  $\tan \theta$  = rate of rxn.

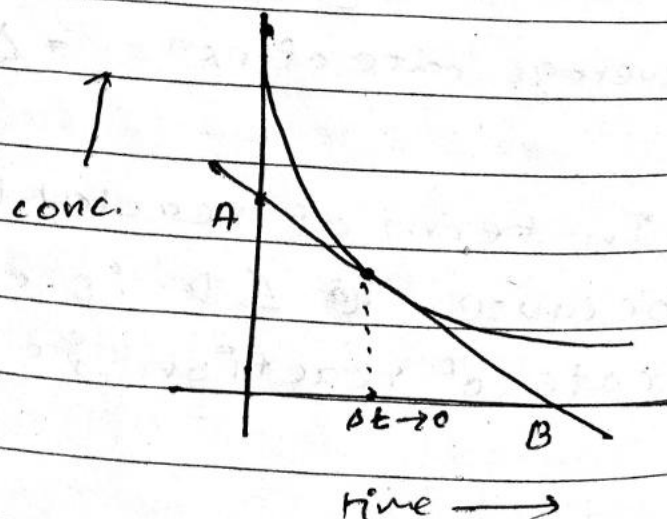
ii) Instantaneous rate of reaction.

→ The <sup>rate of</sup> change in concentration of reactant or product at a instant is called instantaneous rate of reaction.

in, Instantaneous rate,  $\Delta t \rightarrow 0$

$\therefore$  Instantaneous rate of rxn =  $-\frac{\Delta C}{\Delta t}$

Graphically,



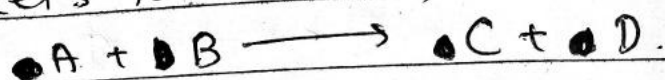


the tangent's slope at a instant gives instantaneous rate of rxn.

or, Instantaneous rate of rxn =  $-\frac{d[D]}{dt}$  at  $t \rightarrow 0$

### (\*) Rate of reaction and stoichiometry.

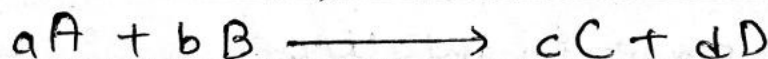
Let's take a rxn,



In above rxn, rate of consumption of A and B is equal to rate of disappearance of C and D. This is only valid for rxn having same stoichiometric coefficient

$$\text{i.e. } -\frac{d[A]}{dt} = -\frac{d[B]}{dt} = \frac{d[C]}{dt} = \frac{d[D]}{dt}$$

another rxn,



→ In this reaction,

$$-\frac{d[A]}{dt} \neq -\frac{d[B]}{dt} \neq \frac{d[C]}{dt} \neq \frac{d[D]}{dt}$$

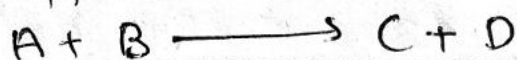
Because their no. of mole are different. So, to make them equivalent to each other, we can rewrite.

$$\therefore -\frac{d[A]}{dt} \times \frac{1}{a} = -\frac{d[B]}{dt} \times \frac{1}{b} = \frac{d[C]}{dt} \times \frac{1}{c} = \frac{d[D]}{dt} \times \frac{1}{d}$$

### (\*) Rate law equation.

→ From Law of Mass action, we know that Rate of reaction is directly proportional to product of molar concentration of reaction raised to certain exponent.

Suppose a reaction,



Then,

$$R \propto [A]^m [B]^n$$

$$\therefore R = k [A]^m [B]^n \text{ ——— (i)}$$

where  $k$  is rate constant. It is defined as rate of reaction ~~when~~ when ~~the~~ the value of concentration of reactant is taken as unity.

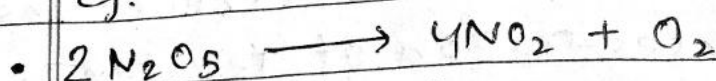
$m$  and  $n$  are exponents which can be found experimentally.

Eqn (i) is called rate law equation which gives relation between molar concentration and rate of reaction.

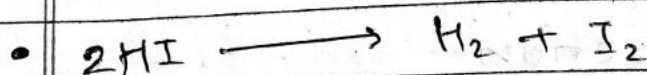
### (\*) Order of reaction.

→ The sum of exponent of reaction in rate law eqn is called order of reaction.

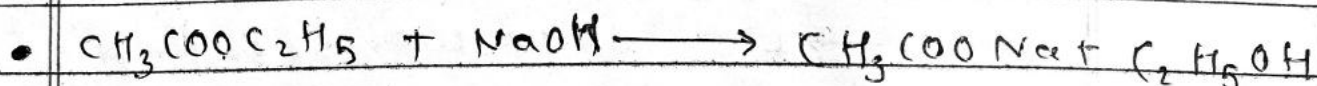
Eg.



$$R = k [\text{N}_2\text{O}_5]^1$$

 $\therefore$  order of rxn = 1

$$R = k [\text{HI}]^2$$

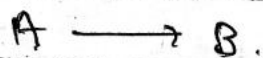
 $\therefore$  order of rxn = 2

$$R = k [\text{CH}_3\text{COOC}_2\text{H}_5]^1 [\text{NaOH}]^1$$

 $\therefore$  order of rxn = 1 + 1 = 2

→ The value of order of reaction may be zero, ~~any~~ +ve integer or fractional also.

① Zero order reaction.



$$- \frac{d[A]}{dt} = R = k [A]^0$$

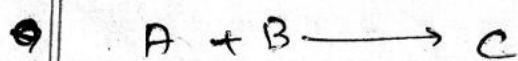
$$\therefore R = k$$

→ The rxn in which the sum of exponent of molar concentration of reactant in rate law eqn is zero.



- The reaction whose rate of rxn is independent of concentration of reactant.  
Eg. Rxn of  $H_2$  and  $Cl_2$  in presence of water and sunlight.

### (i) First order reaction



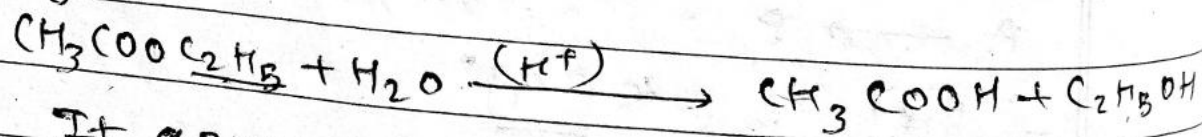
$$R = k[A]^1[B]^0$$

order  $\rightarrow 1$

- The rxn whose order is one.  
→ The rxn whose rate depends on ~~concentr~~ 1st power of concentration of either reactant is called 1st order reaction.

### (ii) Pseudo first order reaction

- The reaction which follows 1st order kinetics but appears to be second order.  
Eg.

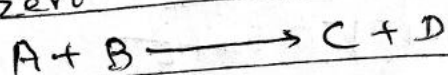


It appears to be 2nd order but experimentally it is found following 1st order kinetics.



\* Unit of Rate constant (K) for different order reaction.

(1) ~~zero~~ order rxn.



$$R = k [A]^0 [B]^0$$

$$k = \frac{R}{[A]^0 [B]^0} = R$$

$$\therefore k = R = \text{mol L}^{-1} \text{time}^{-1}$$

$$\therefore \text{unit of 'k'} = \text{mol L}^{-1} \text{time}^{-1}$$

(2) 1st order rxn.

$$R = k [A]^1 [B]^0$$

$$k = \frac{R}{[A]^1 [B]^0} = \frac{R}{[A]}$$

$$k = \frac{\text{mol L}^{-1} \text{time}^{-1}}{\text{mol L}^{-1}}$$

$$\therefore \text{unit of 'k'} = \text{time}^{-1}$$

(3) 2nd order rxn

$$R = k [A]^1 [B]^1$$

$$k = \frac{R}{[A]^1 [B]^1} = \frac{\text{mol L}^{-1} \text{s}^{-1}}{\text{mol L}^{-1} \times \text{mol L}^{-1}}$$

$$\therefore \text{unit of 'k'} = \text{L mol}^{-1} \text{s}^{-1}$$

iv) 3<sup>rd</sup> order rxn.

$$R = k [A]^2 [B]^1$$

$$k = \frac{R}{[A]^2 [B]^1} = \frac{\text{mol L}^{-1} \text{s}^{-1}}{\text{mol}^2 \text{L}^{-2} \text{mol L}^{-1}}$$

$$\therefore \text{unit of } k = \text{L}^2 \text{mol}^{-2} \text{s}^{-1}$$

(\*) Finding of order of reaction by initial rate method.

Q). The following data was obtained at 300K for the reaction:  $2A + B \longrightarrow C + D$ .

| Exp. No. | [A] mol L <sup>-1</sup> | [B] mol L <sup>-1</sup> | Initial rate (C)      |
|----------|-------------------------|-------------------------|-----------------------|
| 1        | 0.1                     | 0.1                     | $6.0 \times 10^{-3}$  |
| 2        | 0.3                     | 0.2                     | $7.2 \times 10^{-2}$  |
| 3        | 0.3                     | 0.4                     | $2.88 \times 10^{-1}$ |
| 4        | 0.4                     | 0.1                     | $2.4 \times 10^{-2}$  |

i) what is rate law?

ii) Write the order with respect to each reactant and overall order.

iii) Find the unit of overall reaction.

Soln:- order with respect to A = m

" " " " to B = n

Rate law equation will be,

$$\text{Rate} = k [A]^m [B]^n$$

Substituting values from given table,

## (\*) Molecularity of reaction.

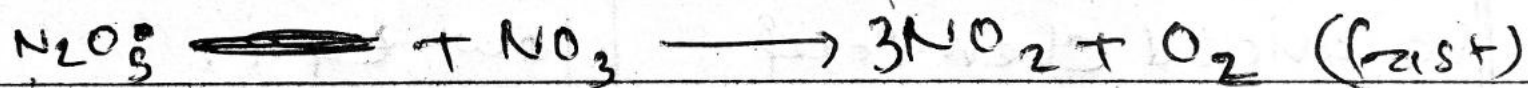
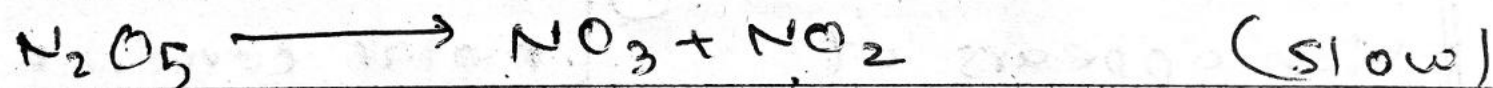
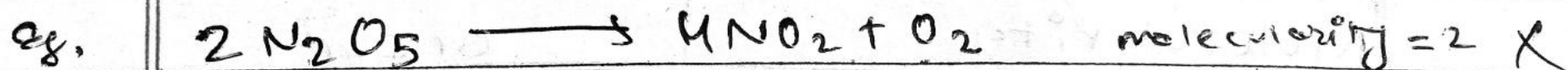
→ It is number of molecules reacting of reactant that appears in balanced chemical reaction.



molecularity = 2

→ In multistep reaction, the molecularity is numbers of molecules reacting in slowest step.





In slow step, the no. of molecules reacting is 1.  $\therefore$  molecularity of overall rxn is 1.

→ The step which product unstable product first is referred to be slowest step.

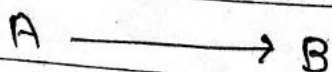
| Molecularity                                                        | Order                                             |
|---------------------------------------------------------------------|---------------------------------------------------|
| → No. of molecules reacting which appears at balanced chemical rxn. | → Sum of exponent of molar conc. of rate law eqn. |
| → Theoretical                                                       | → experimental                                    |
| → always whole no.                                                  | → may be fractional                               |
| → never zero                                                        | → may be zero                                     |
| → doesn't vary                                                      | → vary when changes in temp, conc. occurs.        |

### \* Integrated rate equation.

→ Eqn which gives relation between time, rate constant and concentration.

#### ① Zero order reaction.

Let initial concentration of reactant at  $t=0$  is  $a$ . 'x' concentration of product is formed at  $t=t$ .



$t=0$      $a$                        $0$

$t=t$      $a-x$                        $x$

Now, the rate of reaction at time 't' is

$$-\frac{d[A]}{dt} = k[A]^0$$

$$\text{or, } -\frac{d(a-x)}{dt} = k$$

$$\text{or, } -\frac{da}{dt} + \frac{dx}{dt} = k$$

$$\text{or, } dx = k \cdot dt$$

Integrating both sides,

$$\int dx = k \int dt$$

$$\text{or, } x = kt + C$$

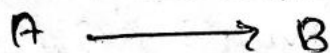
where  $C = \text{integration constant}$

$$x=0, t=0,$$

$$C=0$$

$$\therefore x = kt$$

(ii) 1st order reaction.



$$t=0 \quad a \quad 0$$

$$t=t \quad a-x \quad x$$

Now, rate of eq<sup>n</sup>,

$$-\frac{d[A]}{dt} = k \cdot [A]$$

$$\text{or, } -\frac{d(a-x)}{dt} = k(a-x)$$



$$\text{or, } \frac{dx}{dt} = k(a-x)$$

$$\text{or, } \frac{dx}{(a-x)} = k \cdot dt$$

Integrating both side,

$$\int \frac{dx}{(a-x)} = k \int dt$$

$$\text{or, } \int \frac{dx}{(a-x)} = kt$$

$$\text{or, } -\ln(a-x) = kt + I$$

where  $I = \text{integration constant.}$

when  $t=0, x=0$

$$\cancel{-\ln(a-x)} - \ln(a-0) = I$$

$$\therefore I = -\ln a.$$

Hence,

$$-\ln(a-x) = kt - \ln a$$

$$\text{or, } \ln a - \ln(a-x) = kt$$

$$\text{or, } \ln \left( \frac{a}{a-x} \right) = kt$$

$$\therefore kt = 2.303 \log_{10} \left( \frac{a}{a-x} \right)$$

(\*) Half Life of Reaction,

→ It is time ~~consumed~~ required to consume half of the initial concentration of reactant is called half life.

→ It is denoted by  $t_{1/2}$  or  $t_{0.5}$

If  $a$  = initial concentration,  
amount of reactant left after one half life =  $\frac{a}{2}$

• after two half life =  $\frac{1}{2} \times \frac{a}{2} = \frac{a}{2^2}$

⋮

∴ after 'n' half life =  $\frac{a}{2^n}$  #

① ~~first~~ zero order reaction

from integrated rate equation,

$$x = kt$$

for half life,  $t = t_{1/2}$ ,  $x = \frac{a}{2}$

$$\frac{a}{2} = k \cdot t_{1/2}$$

$$\therefore t_{1/2} = \frac{a}{2k}$$

(ii) 1<sup>st</sup> order reaction.

from integrated rate eqn,

$$kt = 2.303 \log_{10} \frac{a}{a-x}$$

for half life,

$$t = t_{1/2}$$

$$x = \frac{a}{2}$$

$$\text{or, } k \cdot t_{1/2} = (2.303) \log_{10} \frac{a}{a - \frac{a}{2}}$$

$$\text{or, } t_{1/2} = \frac{2.303}{k} \log_{10} \frac{2x}{x}$$

$$\text{or, } t_{1/2} = \frac{2.303}{k} \log_{10} 2$$

$$\therefore t_{1/2} = \frac{0.693}{k}$$

we have,

$$kt = 2.303 \log_{10} \frac{a}{a-x}$$

$$\text{or, } kt = 2.303 [\log a - \log(a-x)]$$

$$\text{or, } \frac{kt}{2.303} = \log a - \log(a-x)$$



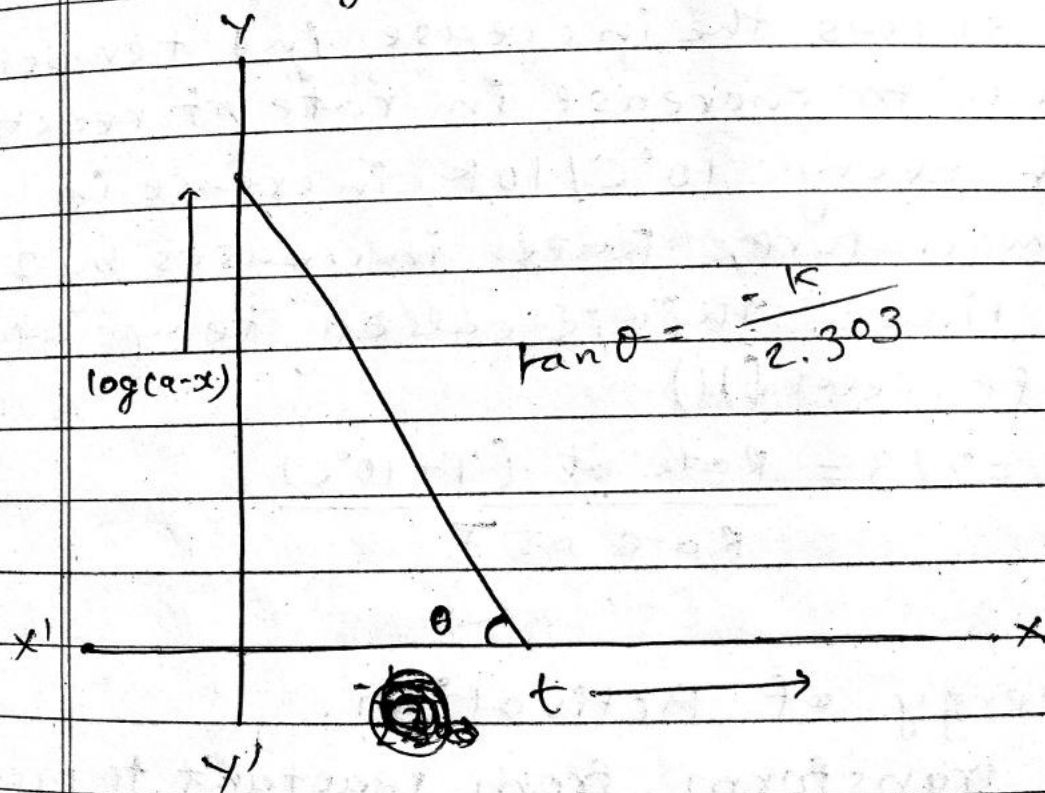
$$\text{or, } \log(a-x) = \log a - \frac{kt}{2.303}$$

comparing it with  $y = mx + c$ .

$$y = \log(a-x)$$

$$m = \text{slope} = -\frac{k}{2.303} \quad x = t$$

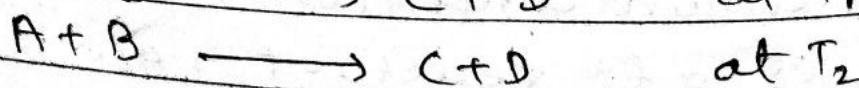
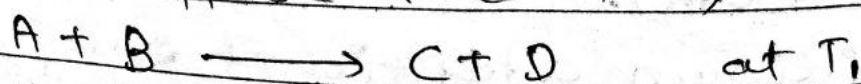
$$c = \log a.$$



$$\tan \theta = \frac{-k}{2.303}$$

(\*) Temperature dependence on rate of reaction.

Let suppose a reaction,



If the molar concentration of reactants is same i.e.  $[A]$  and  $[B]$  in both reaction, then,

$$R_1 = K_1 [A][B] \quad \text{for } T_1$$

$$R_2 = K_2 [A][B] \quad \text{for } T_2$$

If  $T_2 > T_1$ , then,  $K_2 > K_1$ , hence

$$R_2 > R_1$$

→ It shows the increase in temperature leads to increase in rate of reaction.

→ For every  $10^\circ\text{C}/10\text{K}$  increase in Temperature, Rate increases by 2 to 3 times. This is called temperature coefficient ( $M$ )

$$M = 2/3 = \frac{\text{Rate at } (T+10^\circ\text{C})}{\text{Rate at } T}$$

↑  
energy

coefficient ( $\mu$ )

$$\mu = 2/3 = \frac{\text{Rate at } (T+10^{\circ}\text{C})}{\text{Rate at } T}$$

### ⑧ Energy of Activation.

→ To transform from reactant to product the reactant must cross Energy barrier.

The highest point of energy barrier is called ~~activation~~ threshold

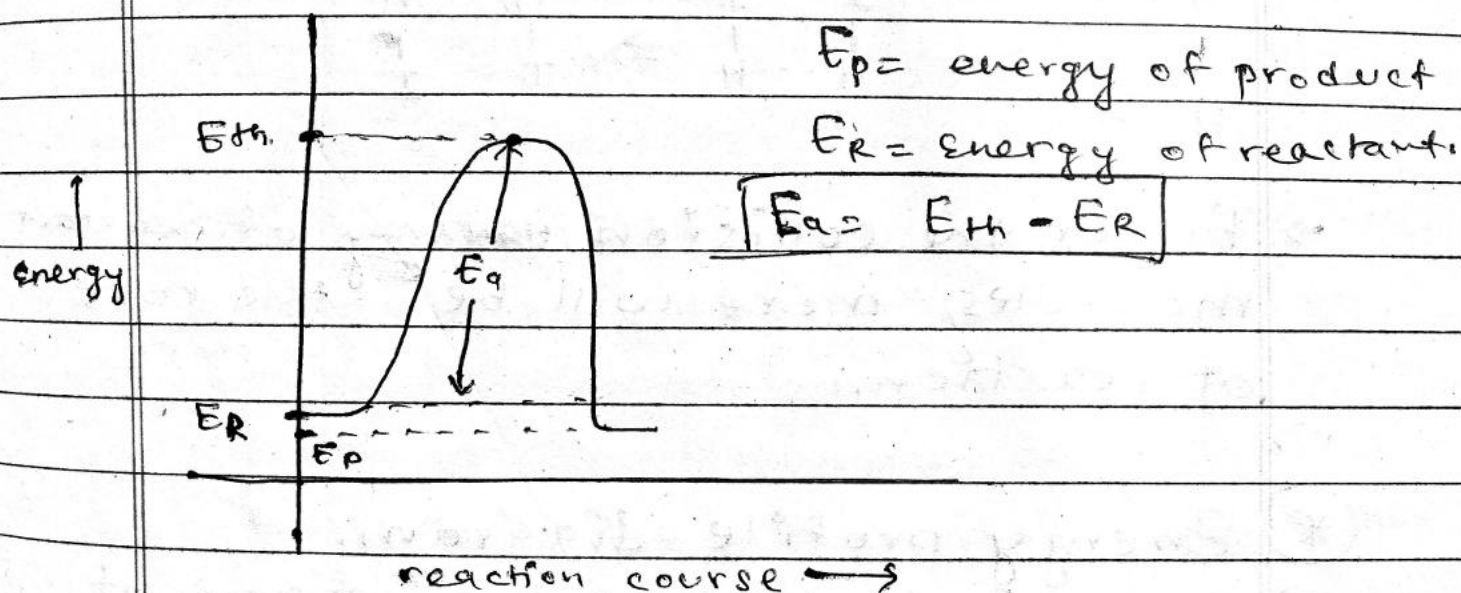
Energy ( $E_{th}$ ).

→ The minimum additional energy that molecules of reactant have to gain



in order to cross energy barrier is called activation energy ( $E_a$ )

→ Activated ~~state~~<sup>complex</sup> is the state where the reactant molecules have crossed the threshold energy and old bonds are breaking and formation of new bond is occurring.

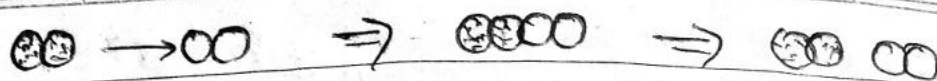


(\*) Collision theory for rate of Reaction.

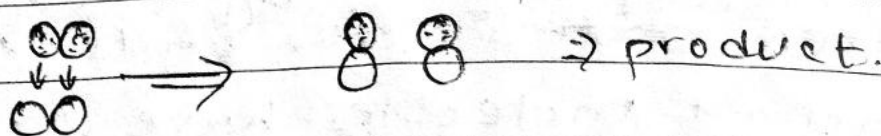
→ The molecules of reactant must collide for reaction to occur.

→ The collision of molecule must be effective collision :-

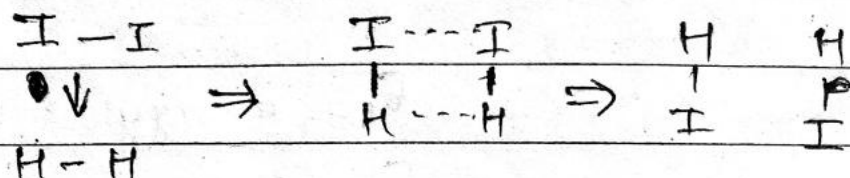
- [ molecules must gain  $E_a$
- [ collision must occur in proper orientation.



No product



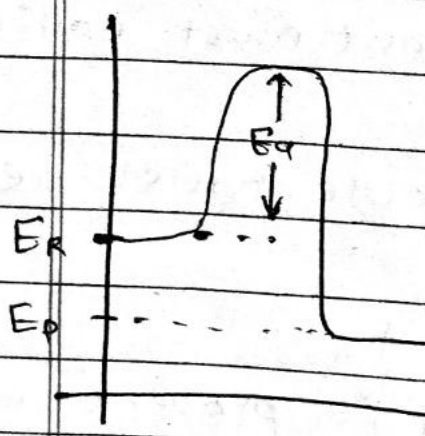
Eg.



→ Faster the collision ~~energy~~ of reactant molecules, more will be the rate of reaction.

(\*) Energy profile diagram.

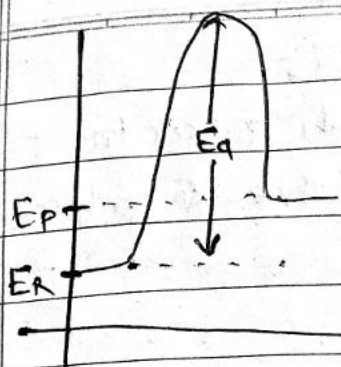
→ Graphical plot between energy of reactant, energy of product with course of reaction.



→ for exothermic reaction

$$E_R > E_P$$

$$\therefore \Delta E = E_P - E_R = -ve$$


$$E_p > E_R$$

$$\therefore \Delta E = E_p - E_R = +ve$$

### Relation between $F_a$ and $R$

- i) when  $E_a$  is more,  $R$  is less.
- ii) When  $E_a$  is less,  $R$  is more
- iii) when  $E_a$  is almost zero,  $R$  is very fast reaction and may be explosive too.

## Catalysis and types of catalysis.

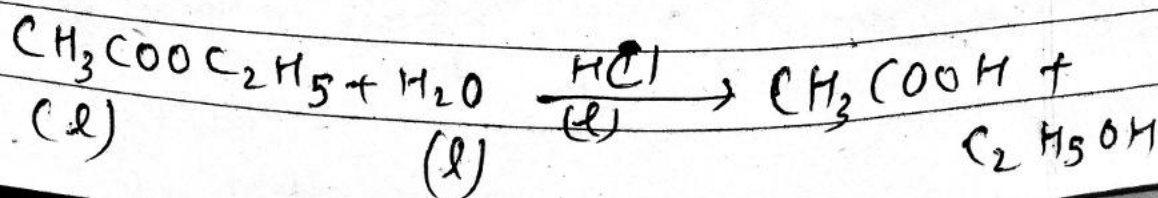
⇒ Catalyst is a substance which alters the rate of chemical reaction without being consumed.

→ The action of catalyst is called catalysis.

Catalysis are of two types:-

① Homogeneous Catalysis

→ The catalysis in which catalyst and reactant are in same phase.

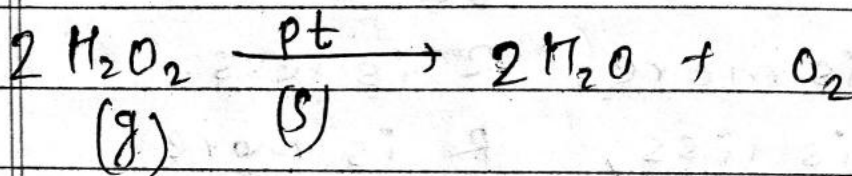
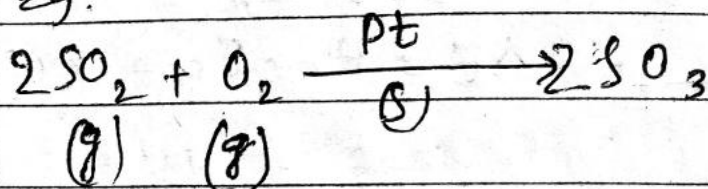




## (2) Heterogeneous catalysis.

→ The catalysis in which reactant and catalyst are in different phase.

Eg.

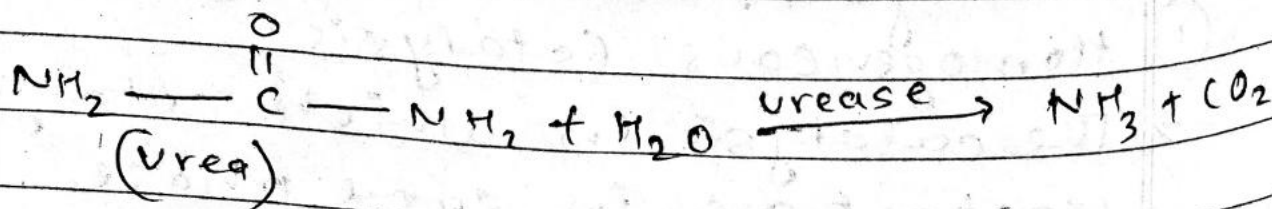
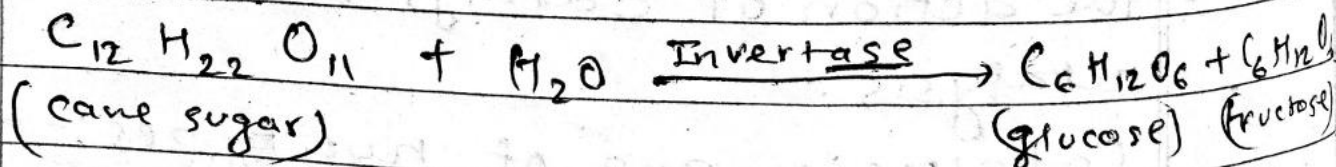


## Enzyme catalysis.

→ Enzyme are complex protein which speed up organic reactions happening in our body.

→ The action of enzyme catalyst is called enzyme catalysis.

Eg.

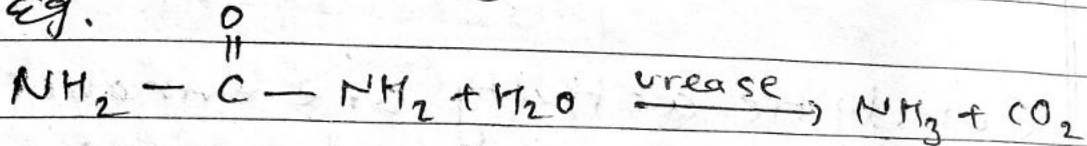


## Characteristics of enzyme catalyst.

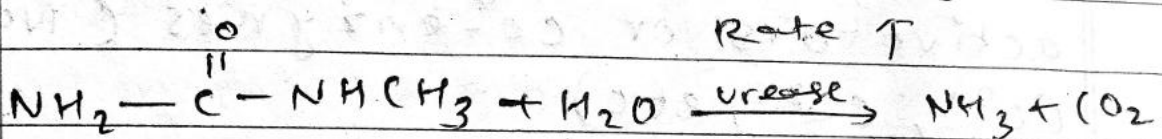
### ① Specificity

→ Particular enzyme catalyses particular reactions only.

Eg.



Rate ↑



Rate ⇒ No change

### ② Rate of reaction catalyzed by enzyme is max<sup>m</sup> at optimum temperature.

→ The action of enzyme is found to be maximum at particular temperature called optimum temperature.

→ For human body, O.T is 37°C (98.6°F)

### ③ Rate of catalyzed reaction is max<sup>m</sup> at optimum pH value.

→ It is found that the action of enzyme is max<sup>m</sup> at particular pH called optimum pH.

→ For human body, O.P is 7.4 (blood)

(4). The catalytic activity is reduced by some substance called inhibitor and completely destroyed by some substance called poison.

(5). Catalytic activity are enhanced by activators or co-enzymes ( $\text{Na}^+$ ,  $\text{Mn}^{2+}$ ,  $\text{Co}^{2+}$ ,  $\text{Cu}^{2+}$ , etc)



## # Factors affecting rate of reaction.

### (i) Nature of reactant

→ Generally inorganic compounds undergoes ionic reaction is fast. Organic compounds undergoes molecular reaction and molecular reaction is slow reaction.

→

### (ii) Concentration.

→ According to Law of mass action, rate of reaction is proportional to product of concentration (Active mass) of reactants raised to power equal to stoichiometric coefficient.

i.e. higher the concentration of reactant, faster will be the reaction.

### (iii) Surface area

→ Larger the surface area of reactant, Larger will be the no. of collisions and faster will be the rxn.

### (iv) Temperature

→ Generally, rate of rxn increases with increase in temperature. On increasing temperature, mobility/velocity of reactant is increased and dissociation of old bonds is also increased which increases

rate of rxn.

→ On increasing temperature by  $10^{\circ}\text{C}$  ( $25^{\circ}$ - $35^{\circ}$ ), rate of reaction is increased by 2-3 times.

(V) pressure.

→ There is no effect of pressure on ~~rate~~ rate of reaction if reactant and product are in solid and liquid. If reactant is in gaseous state, rate of reaction is increased by increasing pressure if volume of ~~decreased~~ product is decreased.

(VI) Catalyst.

→ Catalyst is the substance which changes rate of reaction and itself remains chemically unchanged at the last of reaction.

Generally catalyst increases rate of reaction by decreasing activation energy.

(VII) Radiation.

→ Energy required for some reaction can be provided by light radiation of suitable ~~radiation~~ wavelength.